NLP-953482 Term project proposal

Team members

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Title: Analytic Categories Review System for Course and Instructor Evaluation Result.

Introduction

Course and Instructor evaluation systems are widely used in higher education to collect student feedback for quality assurance and continuous improvement. Traditionally, these systems rely heavily on structured questionnaires with numerical rating scales, which provide quantitative summaries of student satisfaction. While such ratings are useful for statistical analysis, they often lack contextual explanations behind student opinions.

This project proposes an Analytic Categories Review System that applies NLP techniques to analyze student comments together with quantitative scores. The system detects sentiment and categorizes feedback into clear analytical groups to support teaching improvement and academic decision-making

Problem statement

Despite the widespread use of course evaluation systems, several challenges remain:

1.Limited Interpretability of Numerical Scores

Numerical ratings summarize student satisfaction but fail to explain specific causes behind the evaluation results. This limits the usefulness of the data for instructional improvement.

2.Manual Analysis of Textual Feedback

Open-ended comments require manual review, which is:

- Time-consuming
- Labor-intensive
- Difficult to standardize
- Inconsistent across reviewers

3.Lack of Structured Categorization

Existing systems often focus only on overall sentiment analysis (positive/negative) without identifying specific aspects of teaching performance. As a result, institutions cannot systematically identify recurring issues.

4.Inconsistent Interpretation

Manual interpretation introduces subjectivity. Different evaluators may interpret the same comment differently, reducing reliability and fairness.

Objective

- To analyze structured evaluation data using standard statistical methods
- To classify open-ended textual feedback into positive and negative sentiment
- To detect and categorize multiple issues within a single comment
- To organize negative feedback into meaningful subcategories for deeper analysis
- To predict future instructors and courses popularity trends based on historical rating data and sentiment analysis

Motivation

Student feedback is a critical resource for improving teaching quality and learning outcomes. However, the increasing volume of evaluation data makes manual analysis inefficient and unsustainable. Moreover, important details embedded in textual feedback are often overlooked due to time constraints and lack of analytical tools.

The motivation behind this project is to:

- Enhance traditional evaluation systems with deeper analytical capabilities
- Reduce manual workload while maintaining objectivity
- Enable data-driven insights from unstructured text
- Support continuous improvement through structured feedback analysis

System Overview and Methodology

Quantitative Data Analysis

This component focuses on structured evaluation questions, such as:

- Rating-scale questions (e.g., 1–5)
- Multiple-choice or fixed-answer questions

These responses are analyzed using basic data analysis techniques, including:

- Descriptive statistics (mean, distribution, frequency)
- Comparative analysis across evaluation criteria

Trend and Time-Series Analysis

- Post date and student rating
- Use Linear regression technique to forecast the result

NLP-based Qualitative Text Analysis

1.AI Techniques Used

The system applies:

1.1 Supervised Learning

- Sentiment classification (Positive / Negative / Neutral)
- Multi-label subcategory classification
- Possible models: Logistic Regression, Support Vector Machine (SVM), or Transformer-based fine-tuning (e.g., BERT)

1.2 Feature Representation

- TF-IDF vectorization
- Word embeddings (optional advanced approach)

1.3 Rule-Based Support (Hybrid Approach):
Keyword dictionaries are used to strengthen classification consistency within predefined academic subcategories.

2.Sentiment Classification

This task is formulated as a supervised multi-class classification problem.

Labels:

- Positive
- Negative
- Neutral

Possible models:

- Logistic Regression
- Support Vector Machine (SVM)
- Multinomial Naive Bayes

Evaluation Metrics:

- Accuracy
- Precision
- Recall
- F1-score

3.Multi-Issue Detection

A single comment may contain multiple opinions or concerns. The system is designed to identify and separate multiple issues within one textual response. For example, a comment may simultaneously express dissatisfaction with teaching clarity and speaking pace.

4.Subcategory Classification

All feedback comments, regardless of sentiment polarity (positive, negative, or neutral), are categorized into five predefined analytical subcategories.

4.1 Teaching Clarity:

Refers to how clearly the instructor explains content. Includes explanation quality, examples, logical flow, and instructional clarity.

Example keywords:

- Positive → clear, well explained, understandable
- Negative → confusing, unclear, difficult to follow

4.2 Speaking Pace

Refers to delivery speed and verbal presentation. Includes speaking speed, pronunciation, and vocal clarity.

Example keywords:

- Positive → clear voice, appropriate pace
- Negative → too fast, rushed, too slow

4.3 Course Structure

Refers to course organization and design. Includes syllabus clarity, assignment structure, workload balance, and material organization.

Example keywords:

- Positive → organized, structured, well planned
- Negative → messy, disorganized, heavy workload

4.4 Communication Effectiveness

Refers to instructor interaction and responsiveness. Includes answering questions, providing feedback, and encouraging participation.

Example keywords:

- Positive → helpful, responsive, supportive
- Negative → ignores questions, unresponsive

4.5 Professional Behavior

Refers to professionalism and conduct in academic settings. Includes punctuality, fairness, respect, and ethical behavior.

Example keywords:

- Positive → professional, respectful, fair
- Negative → late, rude, unfair grading

Technology and tools

- pandas for data management
- scikit-learn for sense and classification
- NLTK or spaCy for text preprocessing
- FastAPI for communication between frontend and backend
- React for web frontend
- Render for backend deployment (https://nlp-project953482.onrender.com/)
- Vercel for frontend deployment (https://nlp-project953482.vercel.app/)

System Processing Pipeline

1. Dataset preparation
2. Text Preprocessing: Cleaning (remove special characters) Lowercasing Tokenization Stopword removal Lemmatization TF-IDF vectorization
3. Sentiment Classification Using a supervised machine learning model, the system predicts: Positive Negative Example output: Negative
4. Subcategory Classification (Multi-label) The system identifies relevant as: Teaching Clarity, Speaking Pace, Course Structure, Communication Effectiveness, and Professional Behavior
5. Aggregation & Visualization Results are summarized and displayed as: Sentiment distribution charts Subcategory frequency analysis Instructor comparison dashboards

Target Users

- Students – view course and teaching evaluation result
- Instructors – review aggregated feedback for teaching improvement
- Academic Administrators – support academic planning and quality assessment
- Educational Institutions – enhance evaluation systems and quality assurance processes

Core features

- Analysis of both structured and unstructured evaluation data
- Automatic sentiment detection (positive / negative / neutral)
- Multi-label classification for comments containing multiple issues
- Categorization of feedback into meaningful analytical subcategories
- Aggregated reporting and visualization of evaluation results

Milestone

The project will be developed according to the following milestones:

- **Phase 1: Requirement Analysis and Data Preparation**
  Review evaluation system requirements, define analytical categories, and prepare structured and unstructured evaluation datasets.

- **Phase 2: Quantitative Data Analysis Module**
  Implement statistical analysis for structured evaluation data, including descriptive and comparative analysis.

- **Phase 3: NLP Model Development**
  Develop sentiment classification and multi-label category classification models for open-ended textual feedback.

- **Phase 4: System Integration and Testing**
  Integrate quantitative and NLP-based analysis modules and evaluate system performance using sample evaluation data.

- **Phase 5: Result Analysis and Documentation**
  Analyze system outputs, summarize findings, and prepare final documentation and presentation materials.

Literature Review / Business Review

Previous research in higher education evaluation shows that numerical rating systems alone provide limited insight into student learning experiences and teaching effectiveness. As a result, many studies emphasize the importance of analyzing open-ended student feedback to capture contextual and qualitative information.

Recent work in Natural Language Processing highlights the use of sentiment analysis and text classification to automatically analyze large volumes of educational feedback. These approaches have been shown to improve efficiency and consistency compared to manual review. However, existing systems often focus on basic sentiment polarity and lack detailed categorization tailored to academic evaluation.

Expected result

- A functional analytic system that processes both quantitative and qualitative evaluation data
- Clear identification of positive and negative sentiment trends
- Structured categorization of student comments into actionable subcategories
- Improved interpretability of student feedback for academic review and planning
- A predictive model that visualize the popularity trend of instructors and courses for the upcoming academic year

Limitations

- **Dependent on data quality:**
  The accuracy of sentiment and category classification depends on the quality, diversity, and consistency of the training data.
- **Limited understanding of complex language:**
  The system may struggle with sarcasm, irony, or implicit meanings in student comments.
- **Predefined categories:**
  Feedback is classified based on predefined sentiment labels and subcategories, which may not cover all possible issues.

Dataset

[Big Data Set from RateMyProfessor.com for Professors' Teaching Evaluation]

The model uses these columns to train and analyze the rating trend, sentiments, sub-categories, and average quality, and difficulty of the department:

- **professor_name:** name of the professor
- **department_name:** name of the department
- **post_date:** evaluation date
- **star_rating:** average rating of all evaluations from past to recent
- **name_onlines:** name of online course
- **name_not_onlines:** name of onsite course
- **student_star:** rating from each student
- **comments:** comment from student

Training Result

- ### Sentiment Analysis

```
----------------------------------------
EVALUATION
----------------------------------------
Test Accuracy: 0.8856

Classification Report:
              precision    recall  f1-score   support

    negative       0.89      0.74      0.81      1135
    positive       0.88      0.96      0.92      2363

    accuracy                           0.89      3498
   macro avg       0.89      0.85      0.86      3498
weighted avg       0.89      0.89      0.88      3498
```

The sentiment analysis model demonstrates strong overall performance, achieving a **Test Accuracy of 88.56%**.

- **Precision:** The model is highly reliable when making predictions, with a precision of 89% for negative comments and 88% for positive comments. This means that when the model flags a comment as positive or negative, it is correct almost 90% of the time.
- **Recall Insight:** There is a noticeable difference in recall between the two classes. The model is excellent at identifying positive comments (96% recall) but slightly less sensitive to negative comments (74% recall). This indicates that while the model rarely misses a positive review, it might occasionally overlook subtle negative feedback.

- **Sub-Categories Classification**

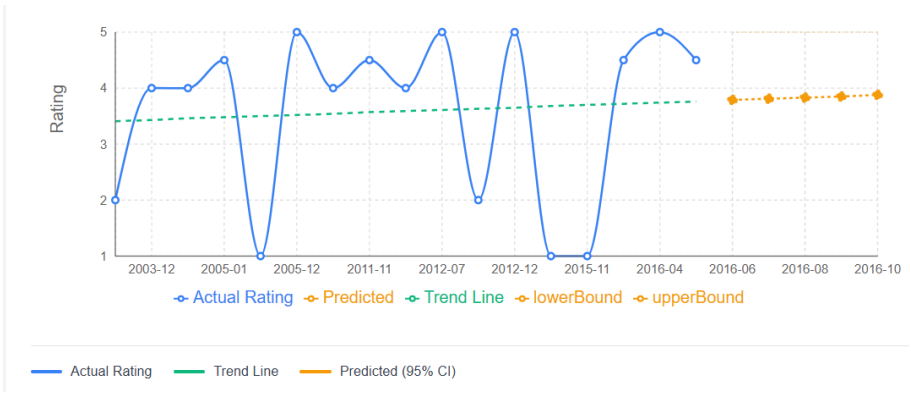
```
----------------------------------------
EVALUATION
----------------------------------------
Training Exact Match Accuracy : 0.9835
Test Exact Match Accuracy     : 0.9700
Test Hamming Loss             : 0.0300
```

	precision	recall	f1-score	support
communication	1.00	0.98	0.99	989
course_structure	0.99	0.92	0.95	1854
professional_behavior	1.00	0.92	0.96	1779
speaking_pace	0.69	0.74	0.72	39
teaching_clarity	1.00	0.93	0.96	1681
micro avg	0.99	0.93	0.96	6342
macro avg	0.93	0.90	0.92	6342
weighted avg	0.99	0.93	0.96	6342
samples avg	0.99	0.94	0.96	6342

The categorization model, which tags reviews with specific traits, shows exceptional predictive capabilities.

- **Exact Match Accuracy:** The model achieved a Test Exact Match Accuracy of 97.00% (closely following the 98.35% training accuracy). In multi-label classification, this is a very strict metric; it means that for 97% of the reviews, the model predicted the *entire exact combination* of tags perfectly without missing a single one or adding an incorrect one.
- **Hamming Loss:** The Test Hamming Loss is remarkably low at 0.0300. This metric evaluates the fraction of individual labels that are incorrectly predicted. A score of 0.03 means that across all tags predicted, the error rate is only 3%.
- **Class-Specific Performance:** The model performs flawlessly across major categories like communication, course_structure, and teaching_clarity, with F1-scores of 0.95 and above.
- **Data Imbalance Observation:** The only area with lower performance is the speaking_pace category (Precision: 69%, Recall: 74%). However, looking at the "support" column, this category only had 39 samples in the test set, compared to thousands for the others. This lower score is a standard result of class imbalance rather than a critical model failure.

- **Trend Analysis**



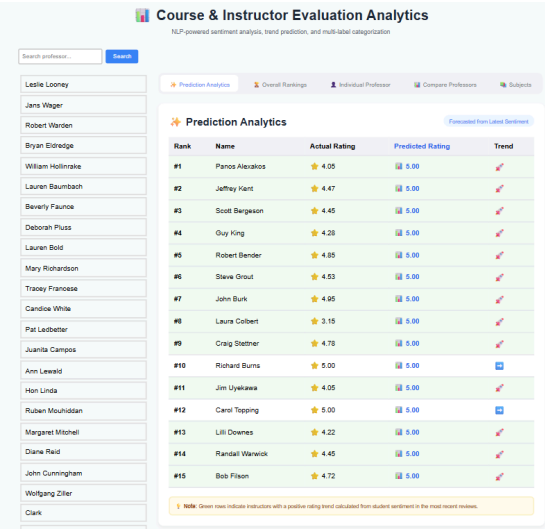
Example of trend analysis for Candice White

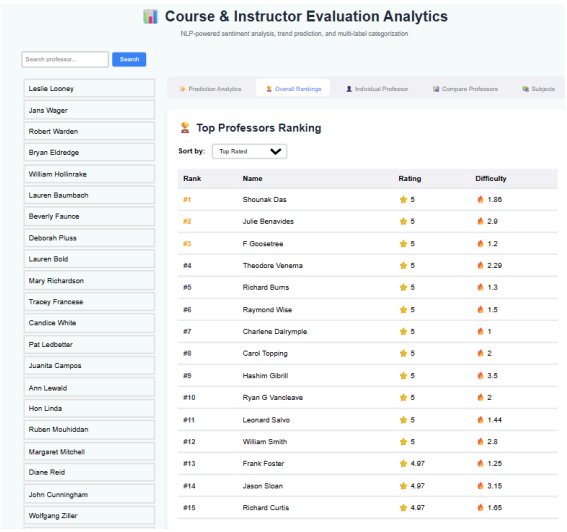
The Trend Analysis module utilizes a Linear Regression model to track an instructor's historical performance and forecast their future ratings.

- **Historical Data Aggregation (Actual Rating):** The solid blue line represents the instructor's actual historical performance. To create a readable time-series graph, the system processes raw student reviews by grouping them into monthly intervals (year_month) and calculating the average rating for each month. This aggregation filters out daily noise and reveals long-term patterns.
- **Mathematical Trend Line Generation:** The green dashed line represents the overarching trend. The system calculates this by fitting a Linear Regression equation to the historical monthly averages. The algorithm evaluates the slope (coefficient) of this line to automatically classify the trend direction: a slope greater than 0.05 is flagged as "increasing," less than -0.05 is "decreasing," and anything in between is considered "stable".
- **Future Forecasting (Predicted Ratings):** The orange dotted line shows the forecasted ratings for future months. The system generates these by extending the linear regression model into the future by a specified number of periods (months).
- **Confidence Intervals (95% CI):** Because machine learning predictions carry uncertainty, the system calculates a 95% Confidence Interval around the forecasted ratings. It computes the standard error of the historical residuals and applies a 1.96 multiplier to create the upper and lower bounds. In Candice White's case, her historical ratings are highly volatile (bouncing between 1 and 5), which is why the model provides a slightly wider confidence interval to account for this historical variance.
- **Logical Constraints (Clipping):** To ensure the predictions make logical sense within a 5-star rating system, the code applies a mathematical clipping function (np.clip). This defensive programming technique guarantees that neither the predicted ratings nor the confidence bounds will ever drop below 1.0 or exceed 5.0.

System User Interface and Data Visualization

- **Global Rankings and Predictive Leaderboards**



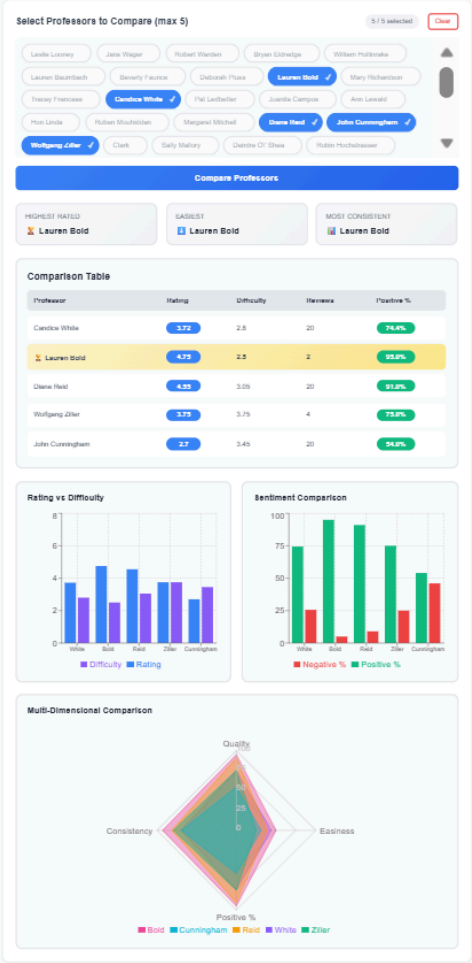


The system provides a macroscopic view of instructor performance through two primary leaderboard interfaces. The "Overall Rankings" dashboard allows users to sort instructors based on historical metrics, such as their average actual rating and perceived difficulty score. Building upon this, the "Prediction Analytics" dashboard introduces predictive sorting. This specialized view leverages the backend Linear Regression model to forecast future performance. It utilizes conditional UI rendering—specifically, light-green row highlighting and directional trend icons (e.g., rockets for positive trends)—to instantly communicate rising performance trajectories derived from recent student sentiment.

- **Detailed Instructor Profiling**

To provide deep insights into individual performance, the system features a granular profiling dashboard, exemplified by the Candice White profile view. This view serves as a visual synthesis of the system's core analytical models. It integrates a time-series forecasting graph, displaying historical rating fluctuations alongside predicted future trends and their 95% Confidence Intervals. Furthermore, it translates the Natural Language Processing (NLP) outputs into easily interpretable visualizations, including a bar chart for sentiment distribution (positive, neutral, negative) and a categorical breakdown of specific pedagogical traits, such as course structure and teaching clarity.

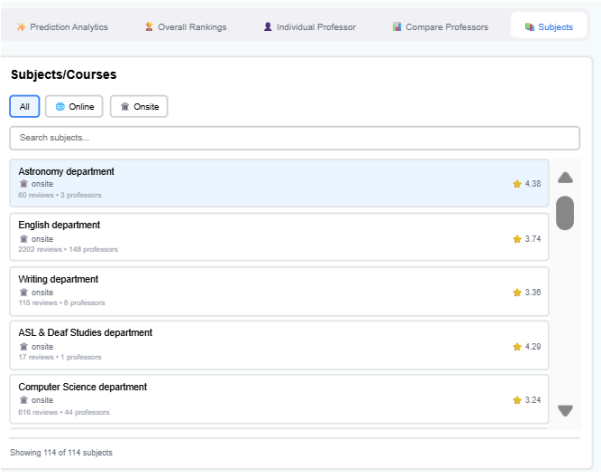
- **Multi-Dimensional comparison**

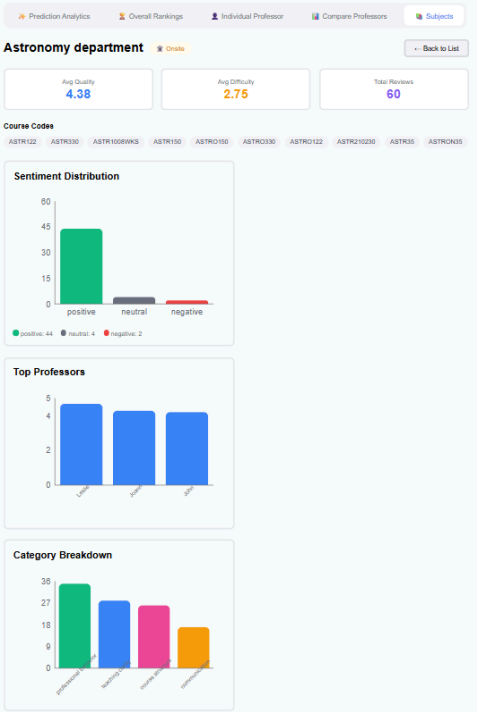


To facilitate data-driven decision-making, the application includes a side-by-side comparison tool capable of evaluating up to five selected professors simultaneously. The system automatically computes and highlights key metrics, dynamically identifying the "Highest Rated," "Easiest," and "Most Consistent" instructors within the user's selected cohort.

A prominent feature of this interface is the Multi-Dimensional Comparison radar chart, which overlays multiple variables, such as Quality, Easiness, Consistency, and Positive Sentiment percentage, allowing users to visually identify the relative strengths and weaknesses of different instructors at a single glance.

- **Department-Level Aggregation**





Recognizing that educational quality extends beyond individual instructors, the system scales its analytical capabilities to the departmental level. The "Subjects" directory provides an aggregated overview of all departments, displaying total reviews, professor counts, and overall average ratings. Upon selecting a specific department (e.g., the Astronomy department), the system renders a dedicated dashboard that aggregates data across all constituent courses and faculty members. This view compiles the collective sentiment distribution, identifies top-performing professors within the department, and aggregates the multi-label category breakdowns, providing both administrators and students with a holistic view of departmental performance.